The Connected Digital Twin
Group Assignment 1
(Technical Summary)

Course: Digital Twins

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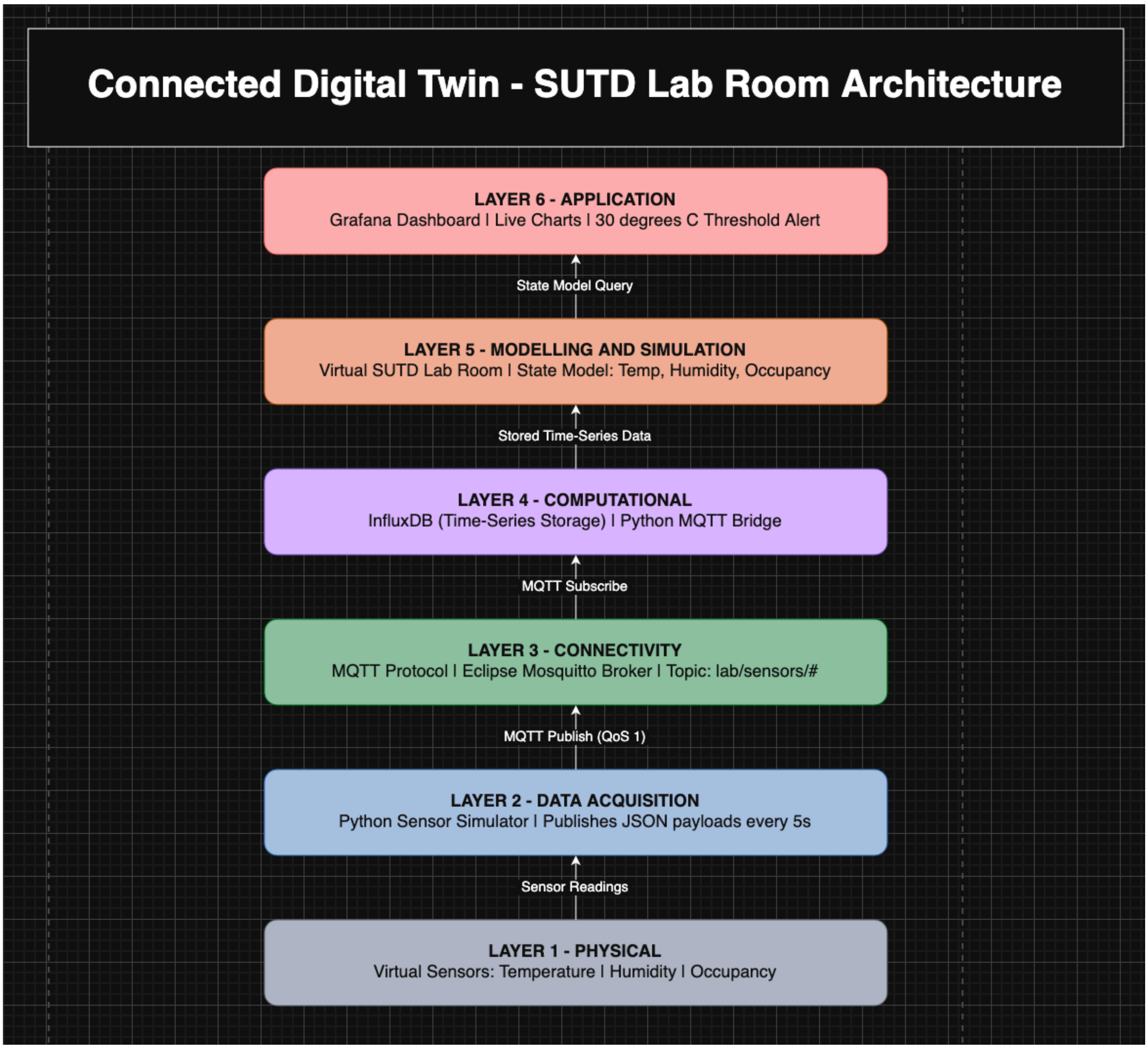
Introduction:
A Digital Twin is a real-time digital replica of a physical object or space, continuously updated with live sensor data to mirror the state of its real-world counterpart. Our project implements a Connected Digital Twin of a virtual SUTD lab room, establishing an unbroken Digital Thread from simulated sensors through to a live dashboard. Three virtual sensors were defined for this space: temperature, humidity, and occupancy, generating continuous synthetic data streams. The result is a real-time Grafana dashboard displaying live sensor readings, with a visual threshold alert triggered when room temperature exceeds 30 degrees Celsius.

System Architecture (Image of Architecture Diagram):
The system is designed around the 6-layer Digital Twin architecture. The Physical layer represents the virtual SUTD lab room and its three sensors. The Data Acquisition layer consists of a Python simulator that generates realistic sensor readings every 5 seconds. The Connectivity layer uses the MQTT protocol via an Eclipse Mosquitto broker to transport sensor data as JSON payloads. The Computational layer comprises InfluxDB, a time-series database that stores all incoming sensor readings, alongside a Python MQTT bridge that subscribes to the broker and writes data to the database. The Modelling and Simulation layer represents the digital state model of the lab room, capturing a real-time snapshot of temperature, humidity, and occupancy. Finally, the Application layer is a Grafana dashboard that queries InfluxDB and renders live charts with threshold alerting. A full architecture diagram is submitted as a separate deliverable.



Messaging Protocol:
We chose MQTT (Message Queuing Telemetry Transport) as the messaging protocol for this project. MQTT operates on a publish-subscribe model, where the sensor simulator publishes data to topics such as lab/sensors/temperature, and the MQTT bridge subscribes to receive and forward that data to InfluxDB. This decoupling of producer and consumer is well-suited for Digital Twin architectures, where multiple systems may need to consume the same sensor feed independently. MQTT was chosen over HTTP because HTTP requires constant polling, introducing unnecessary overhead for continuous sensor streams. MQTT is also significantly more lightweight than HTTP, using minimal bandwidth per message, which is important for real-world IoT deployments where sensors transmit data frequently. Additionally, MQTT supports Quality of Service levels to guarantee message delivery. This project uses QoS 1, ensuring at least once delivery of every sensor reading to maintain an accurate twin state. MQTT is the industry standard protocol for IoT sensor networks and is natively supported by major cloud platforms including AWS IoT and Azure IoT Hub, which can help make our architecture directly transferable to real physical deployments.

Digital Thread:
The Digital Thread refers to the unbroken flow of data from the physical sensor to the application layer. In this project, every sensor reading is timestamped at the point of generation by the Python simulator, published via MQTT, received by the bridge, and written to InfluxDB with full fidelity. Grafana then queries this data in real time, ensuring the dashboard always reflects the current state of the virtual lab room. This complete and traceable data lineage from sensor to dashboard constitutes the Digital Thread of this twin.

Dashboard and Alerting:
The Application layer is implemented using Grafana, which is also connected directly to InfluxDB as its data source. The dashboard displays three live panels: a temperature time-series chart, a humidity time-series chart, and an occupancy panel showing the current people count. The dashboard is set to auto-refresh every 5 seconds. This ensures that the displayed data reflects the latest sensor readings. A threshold alert is configured on the temperature panel at 30 degrees Celsius. When temperature exceeds this value, the chart line turns red and a visual threshold line marks the breach, providing an immediate indication of an abnormal lab condition.

Future Extensions:
The current implementation is unidirectional, with data flowing from the physical layer up to the application layer. A full bidirectional Digital Twin would introduce a feedback path from the application layer back to physical actuators, for example triggering an air conditioning system automatically when temperature exceeds 30 degrees Celsius. Future iterations could also incorporate data-driven machine learning models to predict sensor behaviour, detect anomalies, and optimise room conditions proactively. These extensions align with the higher layers of Digital Twin maturity, progressing from a Connected Twin toward an Intelligent and eventually Autonomous Twin.

Conclusion:
Our project successfully implements a Connected Digital Twin of a virtual SUTD lab room, fulfilling all four key tasks outlined in the project brief. A synthetic data pipeline was established using Python, MQTT, and InfluxDB, with a live Grafana dashboard providing real-time visibility into room conditions. The system demonstrates a complete 6-layer Digital Twin architecture and an unbroken Digital Thread from sensor to dashboard. MQTT was selected as the messaging protocol for its lightweight design, publish-subscribe model, and industry adoption in IoT deployments. The prototype serves as a strong foundational layer upon which intelligent modelling and autonomous control capabilities can be built in future projects.

Appendix:

Full code can be found at :
https://github.com/EmmanuelKevinS/A1-Journey-of-the-Digital-Twins-Course/tree/main

View recording:
https://drive.google.com/file/d/18GKbYeUFesY1m85kGRNWjR_ZQo9E1hAn/view?usp=sharing

Screenshots of the Grafana Dashboard: